Cheat Sheet: ML in the Regime-Change Project

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1. The 60-second story

Problem. Markets switch between calm and stressed "regimes". Models that assume constant behaviour break when the regime changes. **Idea.** Use *change point detection* (CPD) to find where the statistical behaviour of NIFTY 50 shifts, check the shifts against real crises, compare with an HMM, then test honestly whether regime information *predicts* anything useful. **Answer we found.** CPD finds shifts near real crises far better than chance, but regime flags do *not* improve forecasts of future volatility or drawdowns beyond standard volatility measures and India VIX. A clean negative result, tested without cheating.

2. The data feed and dataset structure

Source. Yahoo Finance via the *yfinance* Python library. Daily *adjusted close* prices, Sep. 2007 to Dec. 2025.

Column	What it is	Coverage
NIFTY	NIFTY 50 index (main series)	2007-09-17 on, 4,486 prices
VIX	India VIX (market-implied volatility)	from 2008-03-03
GOLD	Gold BeES ETF (defensive asset)	from 2009-01-02
LIQ	Liquid BeES ETF (cash proxy)	from 2009-01-02
BANK	NIFTY Bank (downloaded, not used in results)	2007-09-17 on

Raw file: *data/prices_raw.csv*, one row per trading day (index Date, format YYYY-MM-DD), five price columns.

Derived series (all from the NIFTY close P_t):

- **Log-return** $r_t = \ln(P_t/P_{t-1})$: 4,485 values. Used because returns, not prices, are roughly stationary and additive over time. Excess kurtosis is 15.2 (fat tails).
- **21-day realised volatility** $\sigma_t = \sqrt{252}\times$ std of the last 21 returns: how wild the market has recently been, annualised.

The ML dataset (what the models actually see). *data/features.csv*: **791 rows** = one forecast date per week (2010-10-11 to 2025-12-01), 14 columns:

Group	Columns
Identifiers	*date* (forecast date d), *tend* (last date of the target window)
Baseline features	*v21* = trailing 21-day vol, *ew* = EWMA vol (λ=0.94), *vix* = India VIX, all known before d
Regime features	*cpd_ratio* (last CPD segment vol / median vol), *cpd_flag* (1 if ratio $>$ 1.3), *hmm_prob* (HMM probability of the stress state)
Targets (labels)	*y_vol* = log realised vol over the next 21 trading days (regression); *y_dd* = 1 if a drop worse than 5% occurs in the next 21 days (classification)
Extras (not used)	*days_since*, *n_breaks*, *fut_vol*, *fut_mdd* (diagnostics and descriptive tables)

One row means: "standing at the start of this week, here is what we knew (features); here is what happened over the next 21 trading days (targets)." That is the supervised-learning setup: X = knowledge before d, y = the future.

Sizes to remember: 791 total samples → first 150 needed for training → **637 out-of-sample forecasts** (2013-09-23 to 2025-12-01). Out-of-sample drawdown base rate 13.5% (86 positives).

3. The pipeline in six steps

1. **Prepare data:** prices → log-returns → rolling volatility.
2. **Detect change points (unsupervised):** PELT segments the return series where mean or variance shifts.
3. **Label regimes:** a segment is "stress" if its volatility exceeds $1.3\times$ the median volatility.
4. **Validate:** are detected breaks near 9 known crises? Compare against random breaks and against an HMM.
5. **Forecast (supervised):** at each week, use only past data to compute features, fit ridge and logistic regression on past samples, predict the next 21 days.

6. **Judge:** out-of-sample R^2, statistical tests, AUC; plus a trading backtest (regime-switching allocation).

4. What each model is and does

(a) PELT change point detection *(unsupervised; no labels)*

Finds break times $\tau_1 < \cdots < \tau_K$ that minimise $\sum_{\text{segments}} \text{cost}(\text{segment}) + \beta K$.

- *Cost* = how badly one Gaussian (mean, variance) fits a segment. Splitting where behaviour changes lowers the cost.
- *Penalty* βK stops it from cutting everywhere. Bigger β means fewer breaks. We use $\beta = 3 \ln n$ (BIC-style): 25.2 on the full data, 19.9 on 750-day windows.
- *Why PELT:* exact optimum (not greedy) and fast (roughly linear time). Constraints: segments of at least 20 days, breaks on a 5-day grid.
- It is **offline**: it sees the whole series, so it finds breaks *after the fact*.

(b) Hidden Markov Model (HMM) *(unsupervised; the benchmark)*

Assumes the market is secretly in one of 2 hidden states (calm or stress), each with its own Gaussian returns, and the state follows a Markov chain. Fitted by EM (Baum–Welch). The state with higher variance = stress. Feature used: probability the *current* day is in stress. Weakness: you must choose the number of states in advance, and flips often (43 switches vs. 16 CPD breaks).

(c) Ridge regression *(supervised; Task V: forecast next-21-day log volatility)*

Linear model $\hat{y} = b + \mathbf{w}^\top \mathbf{z}$, minimising squared error $+ \alpha \|\mathbf{w}\|^2$ with $\alpha = 1$. The penalty shrinks coefficients, which helps because our features are highly correlated (trailing vol, EWMA vol, VIX all move together) and data is small. Features are standardised using *training data only*.

(d) Logistic regression *(supervised; Task D: will a >5% drawdown happen?)*

Outputs a probability $\hat{p} = \sigma(b + \mathbf{w}^\top \mathbf{z})$, trained by minimising log-loss with an L2 penalty ($C = 1$). Same standardisation.

Why such simple models? Only about 640 weekly forecasts, and overlapping, noisy targets. Deep nets (LSTM etc.) would overfit and are unnecessary: the question is *whether regime features carry information*, and a simple linear model is the fairest test. Hyperparameters were fixed in advance, never tuned, so we cannot have overfit to the test period.

Feature sets (the ablation). F0 = trailing + EWMA vol (baseline). F1 = F0 + CPD features. F2 = F0 + HMM. F3 = F0 + VIX. F4 = everything. Plus VIX-only and CPD+HMM-only. Because each set differs by one addition, the gain or loss from each feature group is measured directly.

5. The ML-rigour concepts (this is what examiners probe)

Concept	Plain explanation and how we handled it
Look-ahead bias	Using future information by accident (e.g. fitting CPD on the full sample, then trading on it). Fix: at every forecast date, CPD and HMM are refit using only the previous 750 returns.
Walk-forward validation	Time series cannot be shuffled for cross-validation. We train on the past and predict the next point, every week, expanding the training set. This mimics real use.
Embargo	Targets look 21 days ahead, so a training sample from last week has a target that is not finished yet. We only train on samples whose target window ended *before* the forecast date.
Standardisation	Features rescaled to mean 0, std 1, using training data only (otherwise test information leaks through the scaler).
Regularisation	Penalty on large weights (ridge / L2). Controls variance when features are correlated and data is small.
Overfitting	Fitting noise. Guarded by simple models, fixed hyperparameters, and judging only on out-of-sample forecasts.
Why log volatility	Volatility is positive and right-skewed; its log is closer to normal, so squared-error regression behaves better.
Baselines	A model is only good relative to something: expanding mean, naive "last vol", and F0. Ideas must beat F0 to count.
R^2_{OOS}	$1 - \frac{\text{SSE of model}}{\text{SSE of reference}}$. Positive means better than the reference, negative means worse. Not the in-sample R^2.
Diebold–Mariano test	Tests whether two forecasts have genuinely different accuracy. We use a HAC (Newey–West) variance because 21-day targets overlap by about 80%, so errors are correlated.
AUC	Probability a random drawdown week gets a higher risk score than a random calm week. 0.5 = coin flip.
Brier score	Mean squared error of the predicted probability; lower is better; compared with the base rate.
Block bootstrap	Resamples chunks (26 weeks) rather than single weeks, keeping time dependence, to get a confidence interval for the AUC gain.
Event validation	Precision and recall of breaks near 9 known crises (window $\pm$20 days), compared with random break placement. It shows *localisation*, not prediction.

6. Numbers to memorise

Detection

PELT on returns: 16 breaks

Recall $\pm$20 days: 56% (random: 14%)

Recall $\pm$40 days: 67% (random: 25%)

HMM: recall 67%, but 43 switches

CPD vs HMM label agreement: 84.1%

Drawdown forecasting

AUC 0.43 to 0.56 (coin-flip level)

All bootstrap CIs for gain include 0

Only 86 positives (low power)

Volatility forecasting (637 OOS)

Baseline F0: R^2 vs mean = 0.333

+CPD: -2.2% vs F0, $p = 0.62$

+HMM: -1.2% vs F0, $p = 0.28$

+VIX: +16.6%, $p = 0.006$ (VIX only +17.8%)

CPD+HMM alone: -41.5%, $p = 0.002$

Allocation backtest (2011–2025, 10 bp costs)

Buy and hold: Sharpe 0.46, max drawdown -38.4%

CPD rule: Sharpe 0.32, max DD -26.1%

HMM rule: Sharpe 0.59, max DD -26.1%

Static 60/40 NIFTY/gold: Sharpe 0.71 (best)

7. Likely questions and short answers

Q: Is this really machine learning?

CPD and the HMM are unsupervised learning (structure discovery with no labels); ridge and logistic regression are supervised learning with a proper train/test protocol. The ML contribution is the careful out-of-sample test of whether discovered structure has predictive value.

Q: What is a regime, and what is a change point?

A regime is a period with stable statistical behaviour (mean, volatility). A change point is where that behaviour shifts.

Q: Why not just use the HMM?

It needs the number of states fixed and a Gaussian assumption. CPD makes no such choice; its tuning knob is the penalty. We

use the HMM as a benchmark, and it agrees with CPD on 84% of days.

Q: How did you choose the penalty? Isn't that tuning?
Fixed a priori by a BIC-style rule $3 \ln n$, not tuned on outcomes. We show a sensitivity plot (84 breaks at β=5 down to 3 at β=160) and say precision and recall depend on it.

Q: Can CPD predict a crisis before it happens?
No, and we say so. A break is detected only after post-shift data accumulates; offline CPD even uses future data. What may be predictable is what *follows* (volatility persists), so we tested that, and regime features added nothing beyond trailing volatility and VIX.

Q: Why is a negative result acceptable?
Because it is honest and informative: the regime flags describe volatility but are redundant for forecasting. The VIX, a market-implied measure, is genuinely informative.

Q: Why did CPD recall look so good with the volatility series?
That series gave 105 breaks, so random placement alone reaches 62% recall. We therefore compare against random baselines and at matched break counts.

Q: How do you know there is no data leakage?
Features use only data before the forecast date; PELT and HMM are refit on trailing windows; scaler and models fit on training data only; the embargo drops training samples whose target window overlaps the forecast date; hyperparameters are fixed.

Q: Why ridge and logistic regression instead of random forests / neural nets?
Small, noisy, overlapping sample; the goal is to test marginal information of regime features, for which an interpretable linear model is the fair, low-variance choice. Non-linear models are listed as future work.

Q: Why does the backtest not beat buy-and-hold on return?
Regime rules cut drawdown (-38% to about -26%) but switching late costs return. The static 60/40 NIFTY/gold mix did better, so much of the gain from "moving to gold" is a gold effect, not timing skill. Consistent with the forecasting result.

Q: What are the main limitations?
Single 15-year history; only 9 hand-picked event dates (approximate, to be verified); 86 drawdown positives clustered in few episodes; overlapping targets; no multiple-testing correction; ETF proxies and Yahoo data; fixed, untuned hyperparameters.

Q: What would you do next?
Online CPD to measure real detection delay; leading indicators (flows, credit spreads, option skew); tuned and non-linear models; formal multiple-testing control; more assets.

8. Do-not-say list (avoid these claims)

- Do *not* say CPD "predicts" crises or gives alpha. It localises shifts after the fact; forecasting skill was not found.
- Do *not* quote the 100% recall on volatility breaks without mentioning 105 breaks and the random baseline.
- Do *not* present event dates as exact: they are approximate until verified against sources.